

Spring 2016 QM 1d)

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1 Correct Solution

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle \quad (1)$$

$$i\hbar \frac{d}{dt} \left[e^{-iAt} e^{iBt} |\psi(0)\rangle \right] = \left[e^{iAt} H(0) e^{-iAt} \right] \left[e^{-iAt} e^{iBt} |\psi(0)\rangle \right] \quad (2)$$

$$i\hbar (-iAe^{-iAt} e^{iBt} + ie^{-iAt} B e^{iBt}) |\psi(0)\rangle = (e^{iAt} H(0) e^{-2iAt} e^{iBt}) |\psi(0)\rangle \quad (3)$$

$$\hbar(Ae^{-iAt} - e^{-iAt} B) = (e^{iAt} H(0) e^{-2iAt}) \quad (4)$$

$$\hbar(A - B) = (e^{2iAt} H(0) e^{-2iAt}) \quad (5)$$

Rearranging for B :

$$\boxed{B = A - \frac{1}{\hbar} (e^{2iAt} H(0) e^{-2iAt}) = A - \frac{1}{\hbar} (e^{iAt} H(t) e^{-iAt})} \quad (6)$$